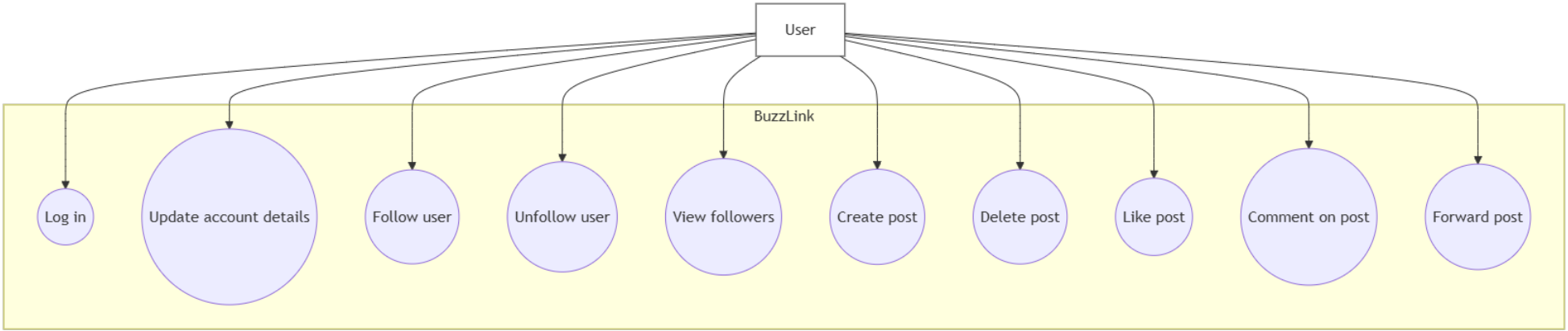
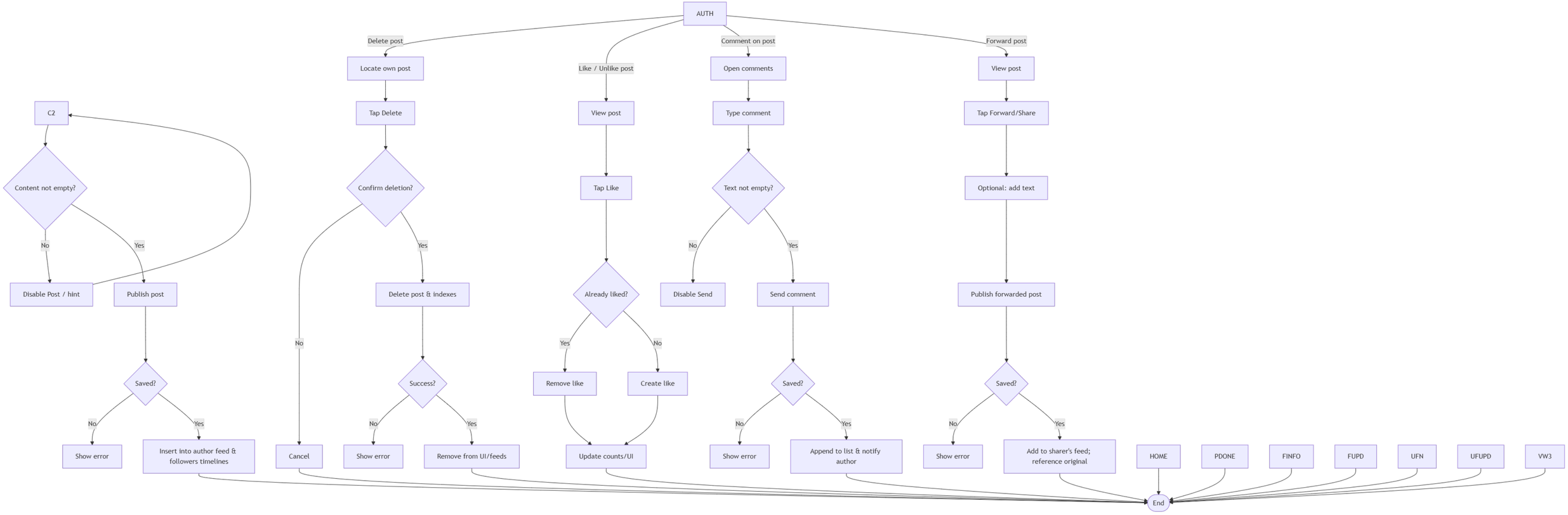
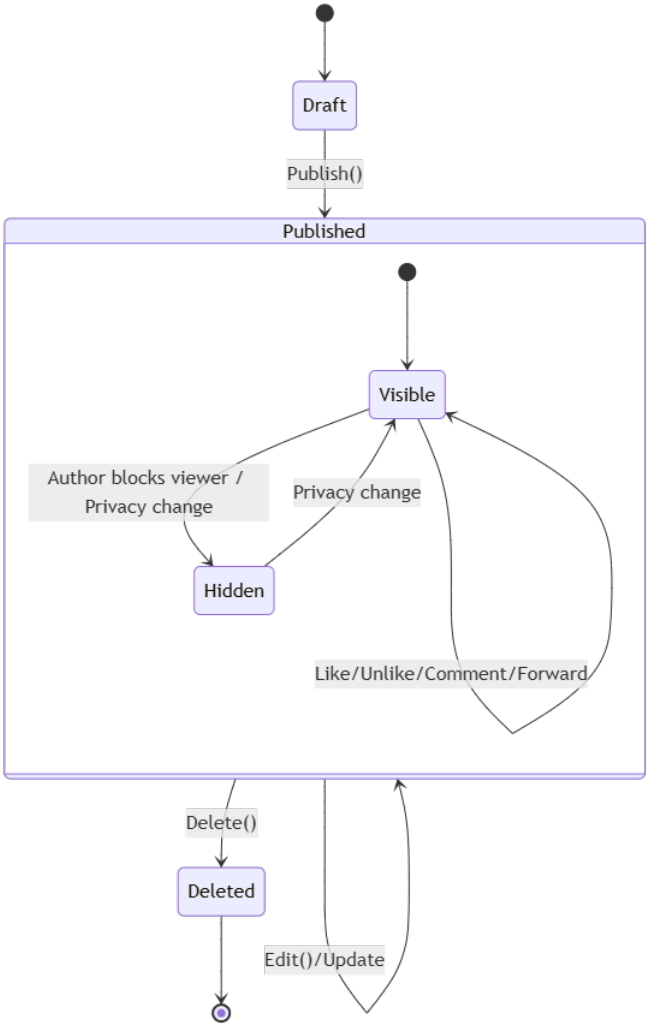
Use case diagram



The unified activity diagram of **BuzzLink** illustrates the complete user workflow within the application. It begins when a user logs in and proceeds through various actions such as updating profile details, following or unfollowing users, viewing followers, and creating, deleting, liking, commenting on, or forwarding posts. Each activity includes validation and system responses like saving data or showing errors. All actions eventually merge back to the main interface, representing a smooth and continuous user experience within the BuzzLink social media platform.

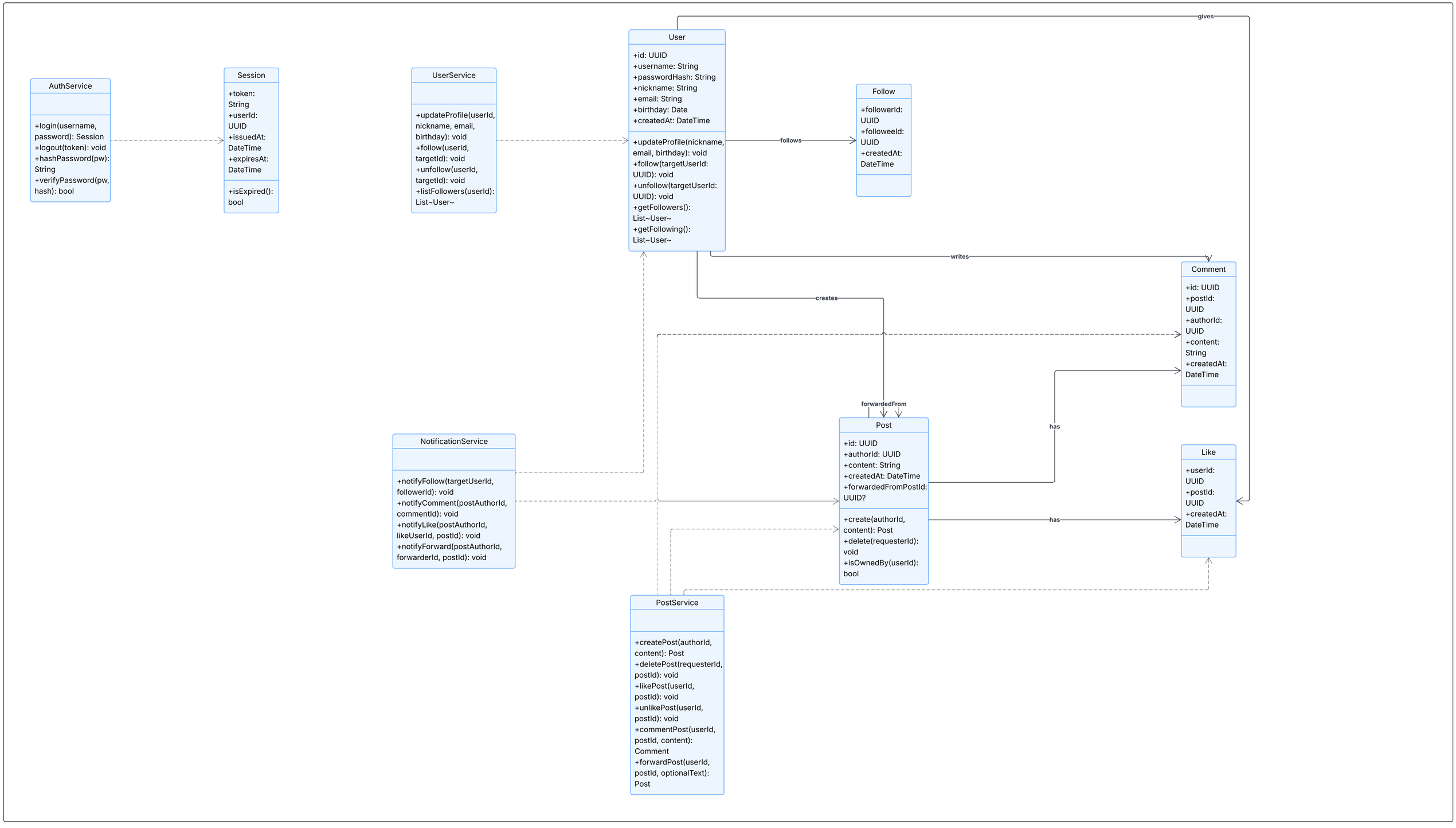
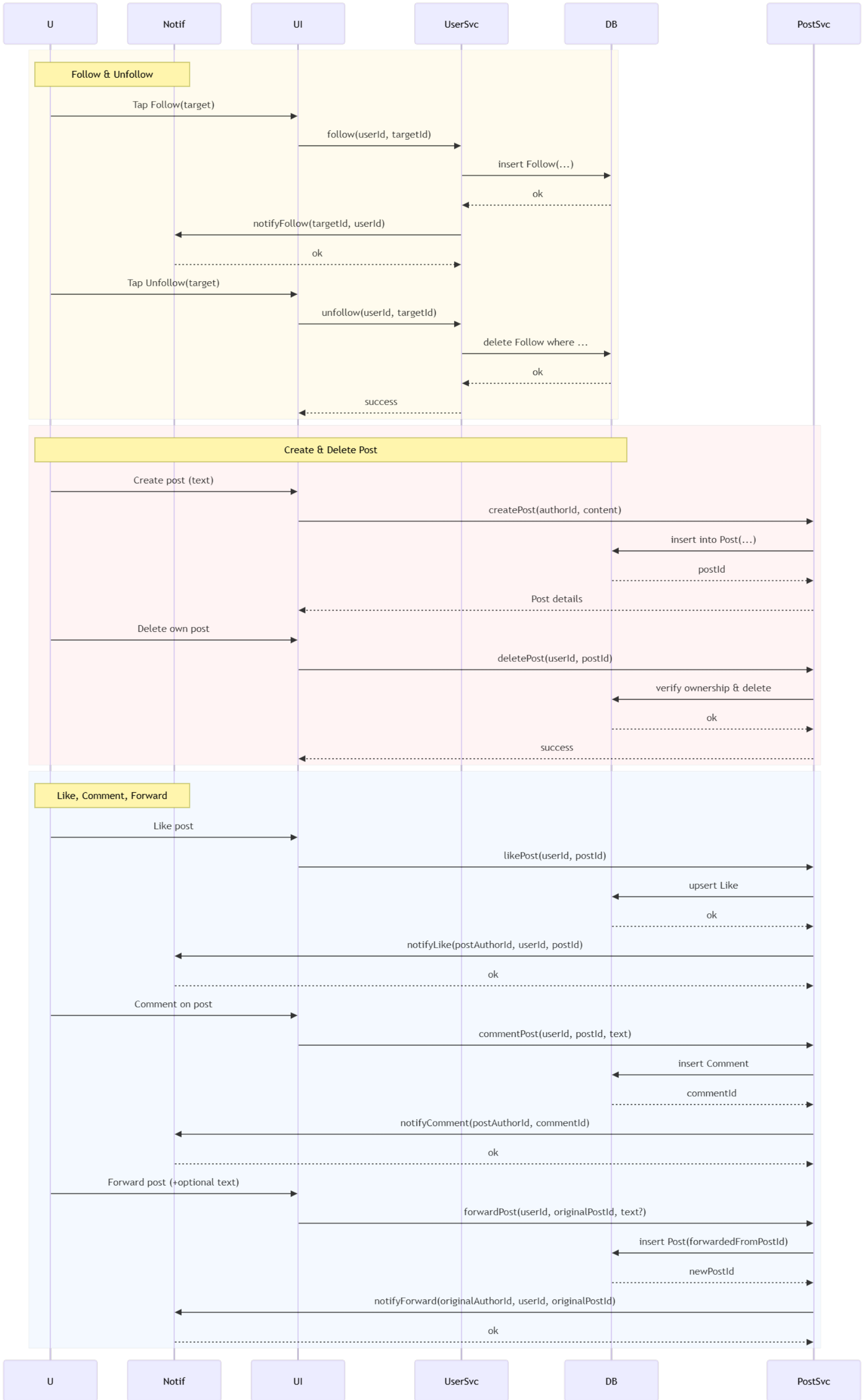


The **State Diagram** for **BuzzLink** represents the lifecycle of a post from creation to deletion. A post begins in the **Draft** state, moves to **Published** once shared, and can later transition to **Deleted** when removed by the user. Within the **Published** state, a post may alternate between **Visible** and **Hidden** based on privacy settings or user actions, while remaining open to interactions such as likes, comments, and forwards. This diagram effectively captures how a post changes status as users interact with it across its entire lifespan.



The **Class Diagram** for **BuzzLink** defines the system's main structure and relationships between components. It includes key classes such as **User**, **Post**, **Comment**, **Like**, and **Follow**, which represent core social media entities. Service classes like **AuthService**, **UserService**, **PostService**, and **NotificationService** manage operations such as authentication, profile updates, posting, and user notifications. Each class contains relevant attributes (e.g., `username`, `content`, `createdAt`) and methods (e.g., `createPost()`, `follow()`, `commentPost()`) to perform specific actions. Together, they form an interconnected model that supports all major BuzzLink functionalities—user management, social interactions, and content sharing.

Social Media Service Class Diagram



The **Sequence Diagram** for **BuzzLink** illustrates how different system components interact to perform user actions in real time. It shows the flow of messages between the **User**, **BuzzLink UI**, and backend services like **AuthService**, **UserService**, **PostService**, and **NotificationService**, all connected to the **Database**. The diagram covers essential interactions such as logging in, updating profiles, following or unfollowing users, creating and deleting posts, and engaging through likes, comments, and forwards. Each sequence highlights how requests are sent, processed, and confirmed, demonstrating BuzzLink's coordinated flow of data and actions across its entire platform.